that it is ready to be written to the memory without any time loss.

Lossless Z-buffer compression allows the reduction of bandwidth sapping Z-buffer traffic by a ratio of 4:1 without any reduction in image quality or precision. There is also a sophisticated visibility subsystem, which eliminates any pixels that are not likely to be visible in a scene. This helps save valuable frame buffer bandwidth, which in turn results in a performance boost.

Last but not the least comes auto precharge, which prepares the memory device by informing it as to which areas of the memory are likely to be used in the immediate future. This means the graphics processing unit doesn't waste any cycles while waiting for the memory—time that is better spent rendering more pixels.

All of these techniques combine to provide a big jump in performance and make visually rich real-time 3D graphics possible, without compromising on performance. The wonder of the GeForce4 is that a tried and tested design is taken to its next level of execution. Faster core cycles and features to match make the GeForce4 almost perfect.

Six flavours in two families

But enough about the intricacies of the NV25 core. What everyone wants to know about is what the cards that will use the chips are going to look like. The desktop GeForce4 line-up is made of six cards that use two different chips. The more powerful of these is the 'Ti' (Titanium) series based on the NV25 core, while the slower versions that use the NV17 core are called the 'MX' series. Within the 'Ti' and 'MX' series too you will find several variants. First up are the thoroughbreds, which will set the pace for performance until nVidia comes out with the NV30 sometime later this year.

1. GeForce4 Ti4600: This card is the top dog and runs at core speeds of 300 to 330 MHz. There is a whopping 128 MB of high speed RAM onboard, which runs at a blazing 650 MHz (128-bit 325 MHz DDR).

2. GeForce4 Ti4400: Using the NV25 core, it will run at a slower 275 MHz with 128 or 64 MB of RAM, depending on the card manufacturer.

3. GeForce4 Ti4200: Powered by the NV25 core, but runs at a more down-to-earth speed of 225 MHz, which is similar to the GeForce3 Ti500. With memory that runs at a slower 500 MHz this card will be much cheaper than the ones mentioned above and is a sure-shot bestseller.

The other members of the Geforce4 family use the NV17 core, which means they lack the pedigree of the Ti4600. But they still pack a mean punch when compared to anything else available today, with the exception of the GeForce3 and the Radeon 7500 and 8500 models.

You won't see realistic reflections and ripples on water with just any old video card

4. GeForce4 MX 460: This one runs at 300 MHz core and will contain 64 MB of RAM running at 550 MHz. The only drawbacks are the slower memory and the lack of advanced features such as pixel and vertex shaders.

5. GeForce4 MX440: This part uses a 270 MHz core and 64 MB of RAM running at 400 MHz.

6. GeForce4 MX420: This is the ultra low-end solution that uses a 200 MHz core and 64 MB of 166 MHz standard SDRAM. This would probably be a successor to the GeForce2 MX200 cards, which are extremely popular today.

The GeForce4 MX has been at the centre of some amount of controversy. Gaming enthusiasts and developers alike have been shouting themselves hoarse that this isn't a real GeForce4. And they have a valid point. The NV17 core sadly lacks the DirectX 8 pixel and vertex shader units. It can process only half as many pixels because it has just two pixel pipelines against the four that are in the NV25 core. This is a serious handicap and implies that the NV17 is not even equal to the GeForce3 in many ways. The only common features between the GeForce4 Ti series and the GeForce4 MX that are worth mentioning are the Light Memory Architecture II, the Accuview AA engine, the nView support and the nVidia's Shading Rasterizer (NSR), which hails from the GeForce2 days.

So where are the games?

There is little doubt that the GeForce4 is brilliant in design and philosophy. But sadly enough the game development industry is moving at a pace that is a lot slower than what the graphics and PC hardware industry has set over the last few years. For instance, while the GeForce3 was released more than a year ago even today there are just a handful of games that can take advantage of its power. Games such as *Medal of Honor: Allied Assault*, *Ballistics*, *Aquanox* and *Return to Castle Wolfenstein* are among the handful that take advantage of all this new technology.

The real fun however will begin when games such as *DOOM III* and *Quake IV* make an appearance sometime next year. Nevertheless, the GeForce4 does allow you to take your present games and turn on every single feature you can think of, with absolutely brilliant image quality and zero 'jaggies' without any performance hit. Definitely something that gamers would consider paying big bucks for!

HATIM KANTAWALLA

Is nVidia's GeForce4 the one for you?

If you're a	Casual gamer	Game enthusiast	Hardcore gamer
And own	A card that runs at 800x600 with 16-bit colour without framing excessively such as a pre-GeForce card—Riva TNT2 or an ATi Rage128	A card that runs at 1024x768 at 32-bit colour with some compromise on texture detail such as a GeForce, GeForce2 MX400 or ATi Radeon 7500	A top of the line card which rocks with all settings pushed to the maximum and anti-aliasing enabled such as a GeForce3 or ATi Radeon 8500
And want to play games like	*Need for Speed III*, *Quake II*, *Half-Life*	*Need for Speed IV*, *Quake III*, *FIFA 2001*, *No One Lives Forever*	*Medal of Honor*, *Return to Castle Wolfenstein*, *Max Payne*
You should	Stick with what you have	Get yourself one of the GeForce4 MX variants	Get a GeForce4 Ti series card and experience complete bliss